



Ministry of Education

Methodica

720



Effective Content for Adaptive Learning

How to Develop Effective Content for AI-Based Adaptive Learning Environments





Gideon Zailer

Adaptive Learning 720

4 principles



Not the Bot Alone

Rich Content → Tailored Path

The Human X-Factor

Metadata:
Know Your Content

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Not The Bot Alone

AI succeeds when it is not
standing on its own

AI

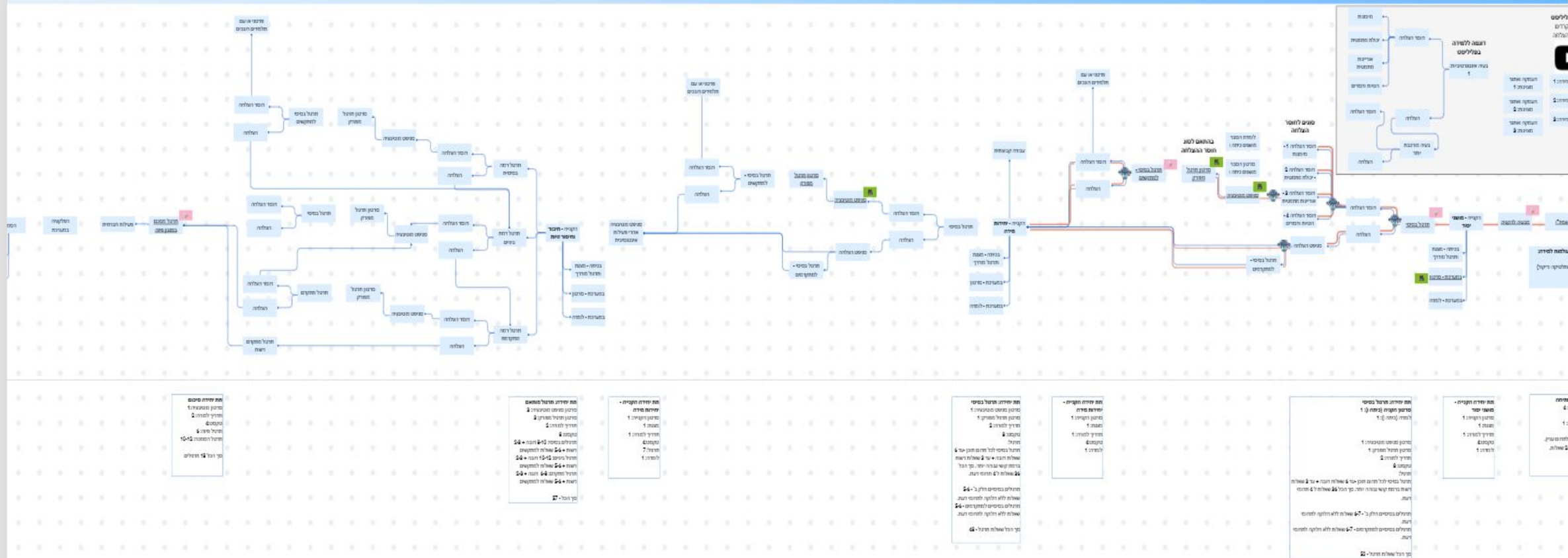
AI agent - orchestrates

CONTENT

High-quality Pedagogy
Carefully designed, partly
automated



A Glimpse of a Learning Journey



Adaptive Learning 720

4 principles



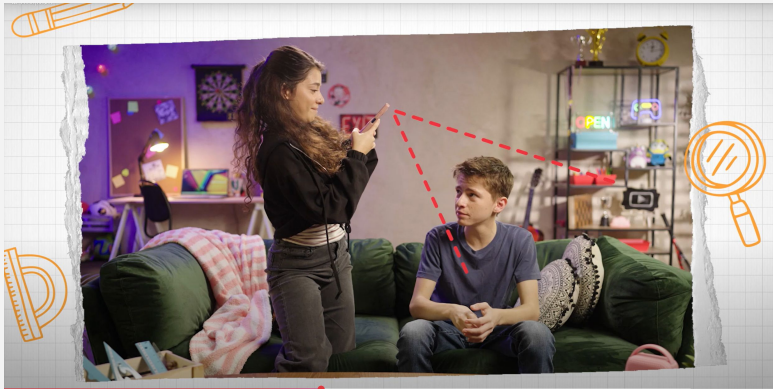
Not the Bot Alone

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The Human X-Factor

Metadata:
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Relevance & Agency



The Human X-Factor

Feeling human in an AI world

Relevance
& Agency

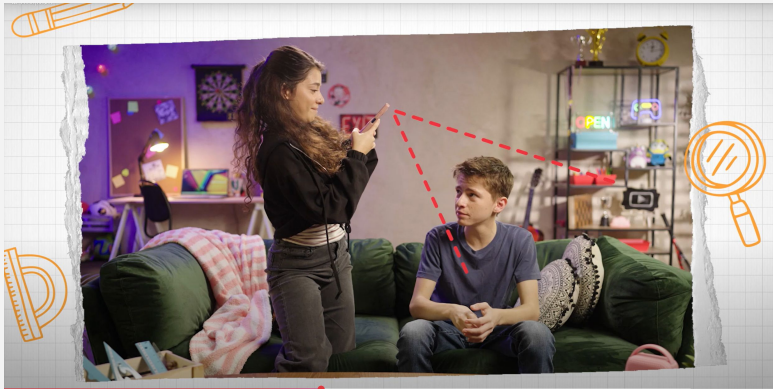
Choice. Choice.
Choice.

Social
Learning

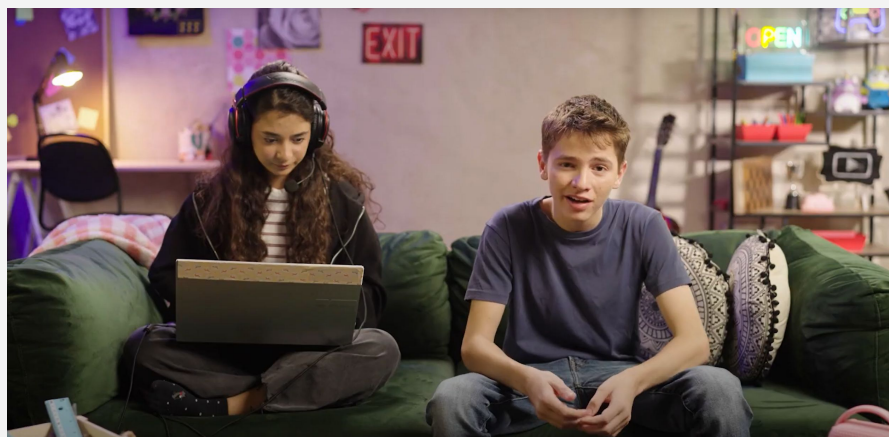
Teacher as
a Mentor



Relevance & Agency



Choice. Choice. Choice.



Social Learning

Looking for angles together

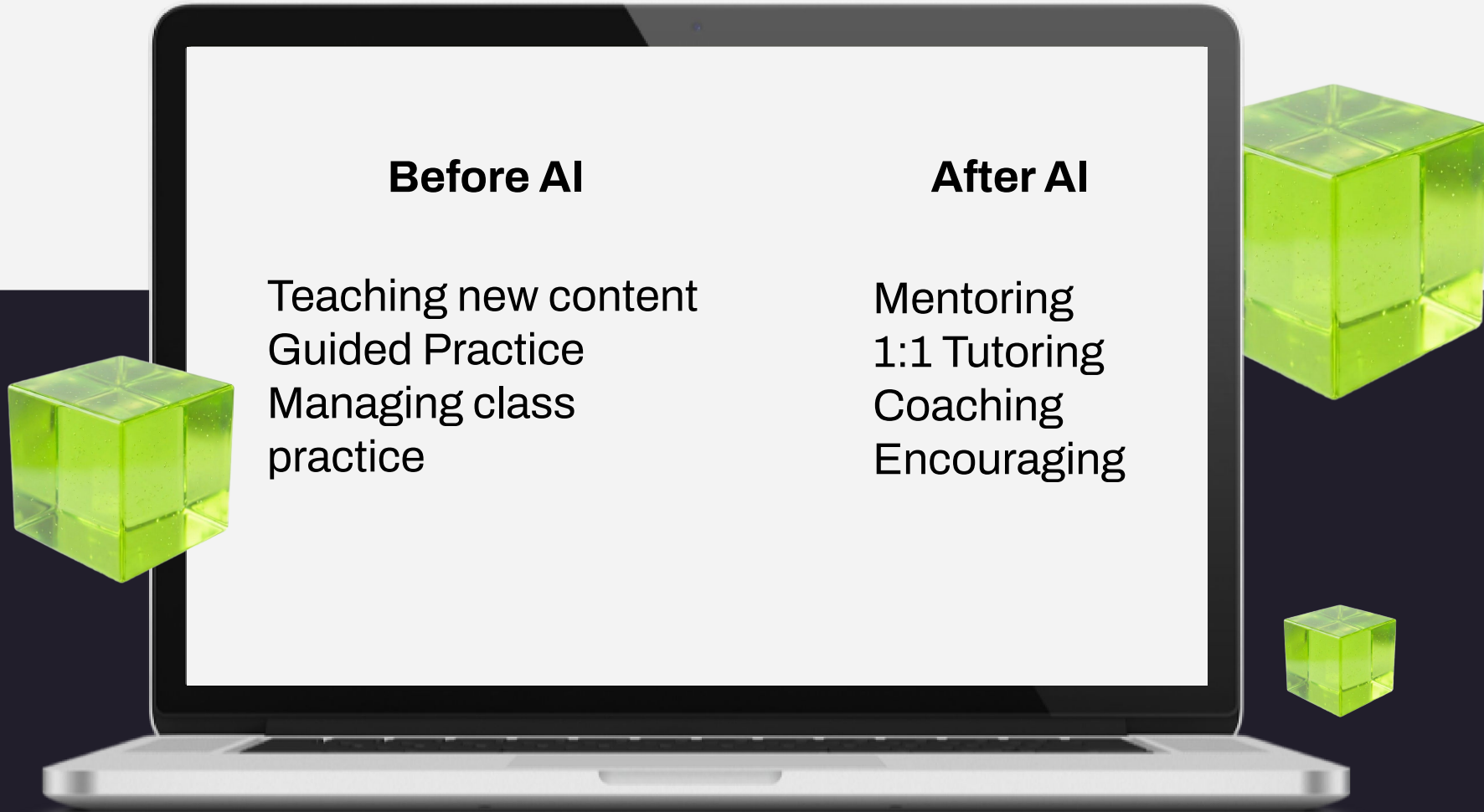
Angles are found everywhere.

Divide into groups of 3-4 students and find:

- 2 **acute angles** in the classroom
- 3 **obtuse angles** in the schoolyard
- 3 **straight angles** on the sports field.



Teacher as a Mentor



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Adaptive learning requires abundance behind the scenes.



Short

Wide

Multiple
variations

Partly
automated

1 Question, 7 Follow-Ups

Roller Coaster

Gravitational potential energy

Kinetic energy

Thermal energy

1. **Trampoline**

2. **Solar Oven**

3. **Advanced Question:**

A solar panel that charges a phone

4. **Struggling Students:**

Mistake 1: Identifying the type of energy

Mistake 2: Classifying as potential / kinetic

Mistake 3: Classifying as transfer / conversion

Mistake 4: Correct representation of change



Adaptive Learning 720

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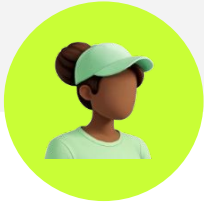


Metadata: Know Your Content

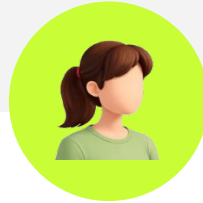
Smart tagging is the gateway to a successful customized learning



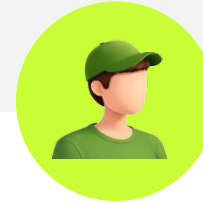
Same Content - 4 People



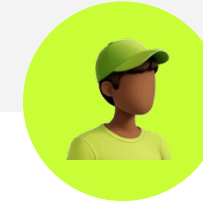
Noa



Sara



Amir



Daniel

Status	Understands the triangle area formula but relies on visual intuition ; struggles with non-standard heights	Strong conceptual understanding; correctly identifies base and height in complex cases	Knows the formula but is confused when base and height are non-standard	Conceptual understanding and critical thinking, but hesitates in non-standard cases
Gaps	Identifying external heights; overcoming visual bias	None – ready for next step	Understanding base–height relationships	Confidence
Next Activity	Interactive drag-and-drop activity showing perpendicular height outside the triangle	Advanced real-world challenge : calculating areas on a soccer field	Short animated explainer reinforcing core concepts	Empowering, inspiring



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Thank you

